

DESIGN AND DEVELOPMENT OF AN EMBEDDED SCIENTIFIC CALCULATOR

USING RK4 ALGORITHM ON ATMEGA328P

Objective & Overview

- Develop a standalone scientific calculator on embedded hardware
- Eliminate dependency on built-in math libraries
- Use numerical approximation (RK4) for function evaluation
- Input via keypad and output displayed on LCD
- Designed for efficient computation on microcontrollers

Core Features

- Basic arithmetic operations
- Trigonometric functions
- Logarithmic & exponential functions
- Power, square root, factorial
- Dual operation modes for flexible usage

Results

- Accurate computation of functions using RK4 method
- Correct evaluation of user input expressions
- Stable and consistent outputs for different inputs
- Successful real-time operation on hardware

Methodology

SOFTWARE

- Implemented in Embedded C
- Evaluates user input expressions

HARDWARE

- ATmega328P
- Matrix keypad (Input)
- 16×2 I2C LCD (Output)
- Custom PCB implementation
- 3D-designed enclosure

Conclusion

This project demonstrates that numerical techniques can be used to perform complex calculations on embedded hardware. The RK4 method enables accurate and reliable results within limited computational resources.

Future work

- Integration of advanced functions (integration, differential solving)
- Optimize computation speed

